Keeping the Dog, Muting the Human: Speech Removal for Dog–Human Play Recordings

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Abstract

Recordings of dogs playing with people contain the people's speech, which must be removed before the recordings can be shared, while the dog vocalizations are the data. We study anonymization on 125 minutes of annotated dog–human social play recorded at the ELTE BARKS Lab (30 sessions, 17 dogs, 581 human sound events, 2,055 dog vocalizations). Privacy is measured by speech recognizers from two model families acting as attackers (Whisper small and large-v3-turbo, NVIDIA Parakeet), and the cost by the dog vocalizations lost, overall and per call type; every model is evaluated on dogs it never saw. Three results. (1) A frame classifier on the frozen Whisper encoder and AudioSet class scores, trained to separate human sounds from dog vocalizations, finds 84% of the human sound frames at a cost of 6% of the dog frames (82% with the call-type weights below), against 72% for an AudioSet-only detector; muting its detections leaves 1.0% of the human events readable (unprocessed: 19.8%; background noise alone: 1.0%) and loses a third less dog sound (95% CI 21–51%). (2) Weighting every dog call type equally in training keeps all 48 barks and 92% of the grunts intact (previously 75% and 66%) at the same privacy. (3) Dog sounds under speech, which muting always loses, can be partly kept: a dog extractor conditioned on Whisper's encoder, used only where a router hears a dog call, returns a fifth of the muted dog sound with no measurable change in privacy for any of the three attackers, and a third with a small privacy cost; the share of dog sound lost falls from 16.4% to 9.8% (8.2%). Adding steady noise with the recording's own room tone, 10 dB above it, where the router hears no dog lets the extractor's output fill *all* the muted time: no dog sound is silenced, and three attackers, including one that first removes the noise, read back no new words (only laughter and backchannels); quiet calls come back at about the level of the added noise. We release the code and a one-command anonymizer.

1 Introduction

Bioacoustic studies of dogs increasingly rely on recordings made during natural interaction with people. In dog–human social play the human partner talks, laughs and calls the dog throughout, so a recording that captures the dog's play pants, growls and barks also captures personal data: what was said, and a voice that identifies the speaker. Before such a dataset can be shared, the speech has to be removed. The dog sounds, however, are the point of the recording, and every second muted by mistake is data lost.

Removing speech from environmental recordings has so far been treated as a voice activity detection problem [1, 2]: detect speech, then delete or mute it. Speaker anonymization in the sense of the VoicePrivacy challenge [3] is a different task: it keeps the words and hides the identity. Here the words themselves must go. Two features of dog–human play make detection hard. First, the dog vocalizations are acoustically close to human non-speech sounds: a play pant is a breathy, forced exhalation [4], not unlike a human laugh or sigh. Second, the relevant cost is not generic false alarms but dog sound lost, so the detector should be judged at a fixed loss of dog vocalizations, not at a fixed false-positive rate on all audio.

This paper makes five contributions. (i) An evaluation protocol for this setting: leave-one-dog-out cross-validation, privacy measured by speech recognizers from two model families acting as attackers, and the cost measured as dog vocalization time and dog call events lost, per call type (Section 4). (ii) A detector that combines a frozen speech encoder (Whisper [5]) with AudioSet class scores [6] and is trained on the trade-off that matters, human sounds against dog vocalizations (Section 3). It mutes about as much speech as an AudioSet-only detector with a third less dog sound lost (Section 5). (iii) Call-type-balanced training, which protects the rare calls (barks, whines, moans) without any privacy cost. (iv) A way to keep part of the dog sound that overlaps speech: a Whisper-conditioned dog extractor, applied only where a dog router hears a dog call (Section 6). (v) Negative results that narrow the design space: subtracting a speech estimate, fusing three microphones for detection, deciding every 20 ms, and dog extractors trained from the spectrogram alone.

2 Data

The recordings were made at the ELTE BARKS Lab in Budapest between March and May 2025, during dog–human social play in a laboratory room: 30 sessions of 17 dogs with their human play partners, 124.7 minutes in total, recorded at 48 kHz with a shotgun (27 sessions) or cardioid microphone. Up to two further microphones recorded in parallel; one of them, a wireless clip-on, is on the dog (pants are about 8 dB relatively louder on it than on the room microphone, human sounds about 4.5 dB, in 25 of 30 sessions). Expert annotators marked 581 human sound events (788 s; speech, laughter, claps and other human sounds) and 2,055 dog vocalizations of ten types. Panting dominates: 1,294 pants make up 3,521 s of the 3,966 s of annotated dog sound

(89%). The other types are rarer: 198 growls, 163 grunts, 48 barks or yelps, 33 moans, 25 sneezes, 13 whines, 10 coughs or woofs, 7 shakes and 264 others.[check with the lab: dataset name and citation, annotation protocol, consent and permission to publish]

3 Method

Frames and features. The audio is resampled to 16 kHz and cut into 0.96 s windows every 0.24 s. Two frozen networks describe each window:

- *EfficientAT mn40* [6], a MobileNet distilled from transformer ensembles on AudioSet [7]. Its convolutional feature map is computed on 10 s chunks (the model saturates on 1 s inputs), averaged over the columns inside the window and passed through its classification head: 527 class scores.
- *Whisper small encoder* [5] (int8 ONNX export [8]). The audio is encoded in 30 s windows every 20 s, each 20 ms frame taken from the window in which it is most central (at least 5 s of context on both sides). The 768-dimensional frames inside the 0.96 s window are averaged. The export provides the first decoder layer's cross-attention keys, a fixed full-rank linear map of the encoder output, which carries the same information for a linear classifier. Whisper's representations are known to retain background-sound information [9]; here the question is whether they separate human sounds from dog sounds better than AudioSet scores.

Classifier and objective. A logistic regression ($C = 0.05$, balanced class weights, standardized inputs) is trained on frames that are at least half covered by a human event (positive) against frames with no human sound that are at least half covered by a dog vocalization (negative). Frames with neither are left out of training: muting background costs nothing the lab analyses, so the classifier should not spend capacity on it. The output is smoothed over 5 frames ($\pm$0.48 s).

Call-type weights. Panting is 89% of the dog time, and only three dogs bark (21, 18 and 7 barks). A model trained without one of them has seen barks of only two other dogs and treats an unfamiliar bark as "not a pant", i.e. possibly a voice. Each dog-only training frame is therefore weighted by $(N/N_t)^\alpha$, where N is the number of dog-only frames and N_t that of its call type, and the dog-only frames keep their total weight. With $\alpha = 1$ every call type weighs the same.

Thresholds and muting. Thresholds are set where a fixed share q of the dog-only frames is flagged (3, 6, 10 or 15%; the levels *light*, *standard* and *strict* use 3, 6 and 10%). Flagged frames are padded by 0.12 s and merged, and these intervals are set to zero in the original file. Everything else stays bit-identical.

4 Evaluation

Protocol. All results are leave-one-dog-out: each of the 17 dogs (with its human partner) is processed by a model trained on the other 16. Thresholds are set on the pooled out-of-fold predictions; for the AudioSet-only detector, choosing them inside the cross-validation instead gave the same results.

Table 1: Frame level, leave-one-dog-out: share of human frames found when 3/6/10/15% of the dog-only frames are flagged, and ROC AUC (human vs dog-only).

input	trained vs	3%	6%	10%	15%	AUC
mn40	all	62.3	72.1	80.0	84.9	.927
mn40	dog	65.6	74.4	82.5	86.7	.938
Whisper	all	70.2	78.6	85.4	90.9	.948
Whisper	dog	72.9	80.7	87.5	92.2	.954
Whisper+mn40	all	71.2	81.2	86.6	90.8	.951
Whisper+mn40	dog	**75.9**	**84.0**	**89.4**	**93.2**	**.963**

Privacy. A speech recognizer plays the attacker on each human event window (the event $\pm$0.3 s, loudness-normalized), before and after anonymization. We report the share of events where it reads two or more words, and the share of the words it read in the original that it can still read after muting (*words read back*). Sound tags such as "(dog barking)" are not counted. Three attackers are used: Whisper small (244M parameters) [5], Whisper large-v3-turbo (809M), and NVIDIA Parakeet TDT 0.6B v3 [10], a Fast-Conformer transducer trained on different data that covers 25 European languages including Hungarian. Parakeet matters because the detector itself uses Whisper's encoder. As a floor, the attackers also read background windows of the same length with no human sound: Whisper small writes 2+ words in 1.0% of them, Parakeet in 4.9% and turbo in 24.5%, so for turbo only *words read back* is meaningful. On background audio, the share of an event's original words that turn up is 0.0% for Whisper small and Parakeet and 1.6% for turbo.

Dog cost. The share of dog-only vocalization time (dog annotated, no human annotated) that is muted, and per call type the share of dog calls that remain *intact* (not a single sample muted). Calls that overlap a human sound are muted by design and reported separately.

5 Results

Frame level. Table 1 compares inputs and training objectives. Whisper's encoder alone beats the AudioSet scores at every dog cost, and the two together are best. Training against dog vocalizations only adds 1–5 points for every input. At 6% of the dog frames, the combined detector finds 84.0% of the human frames, against 72.1% for the previous detector (mn40, trained against all frames). At 3% dog cost it finds more human frames than the previous detector at 6%.

End to end. Table 2 shows the muted recordings as read by Whisper small. The new standard setting leaves 1.0% of the human events readable, the level Whisper reaches on background noise alone, and mutes 8.0% of the dog-only vocalization time. The previous detector needed 12.5% to reach 1.4%. A participant bootstrap (2,000 resamples of the 17 dogs) puts the reduction in dog sound lost at 36% (95% CI 21–51%); the confidence intervals of the privacy differences include zero (readable events: within $\pm$1.6 points;

Table 2: End to end, Whisper small as the attacker (581 events): events with 2+ readable words, original words read back, and dog-only vocalization time muted (all in %).

detector, dog frames flagged	readable	words	dog lost
none (unprocessed)	19.8	100	0
background only (floor)	1.0	0.0	–
VGGish [11], 6%	6.0	19.9	9.0
mn40, 6%	2.2	9.6	7.9
mn40, 10%	1.4	5.6	12.5
mn40, 15%	0.9	4.4	17.9
Whisper+mn40, 3% (light)	1.7	7.8	**4.3**
Whisper+mn40, 6% (standard)	**1.0**	5.2	8.0
Whisper+mn40, 10% (strict)	0.7	3.5	12.3
Whisper+mn40, 15%	0.5	2.6	17.8

Table 3: Dog calls with no human sound on them that stay fully intact at the standard setting (%): AudioSet-only detector (10% of dog frames), Whisper+mn40 (6%), and with call-type weights (6%). Leave-one-dog-out.

call type	events	mn40	W+mn40	+ weights
pant	1,116	71	**82**	**82**
other	243	75	83	**87**
growl	188	90	96	**98**
grunt	159	66	84	**92**
bark/yelp	48	75	67	**100**
moan	32	62	75	**100**
sneeze	24	**92**	83	88
whine	7	14	29	**71**

words read back: within ± 6 points). The same holds at the other two levels (light: -46%, CI 28–61%; strict: -31%, CI 16–45%).

Other attackers. Figure 1 repeats the comparison with all three recognizers. Parakeet, which shares nothing with the detector, reads back 9.2% of its original words after the new standard setting and 11.3% after the previous one, while the dog loss drops from 12.5% to 8.0%. It reads 2+ words in 2.8% and 2.9% of the events, below its own background floor of 4.9% (10 events readable only after the previous setting, 9 only after the new one). Whisper large-v3-turbo reads back 14.9% of its words after the new standard setting and 13.1% after the previous one; at equal dog loss (new strict, 12.3%) it reads back 9.9%. For all three attackers the new detector's curve lies below the previous one: at the same dog loss fewer words are left. On the 29 events where Parakeet and Whisper small agree on at least two words of the original (clearly intelligible speech), one event remains readable after the new standard setting, against three after the previous one. The saving in dog sound therefore comes without a measurable loss of privacy for an attacker that shares nothing with the detector.

Dog calls by type. Because 89% of the dog time is panting, the time-based cost is mostly a panting number. Table 3 counts calls instead. The new detector leaves more calls intact for most types: 82% of the play pants (previously 71%), 84% of the grunts (66%). Barks are the main exception (67% vs 75%, 48 events): the speech encoder appears to respond to the harmonic, voice-like bark; sneezes also lose (83% vs 92%, 24 events). All 16 barks it muted came from one dog. Call-type weights (last column) fix this: every call type keeps as many or more calls than with both earlier detectors, except one sneeze, and privacy does not change (Whisper small: 1.0% readable, 5.2% of the words; Parakeet reads back 7.9% of its words instead of 9.2%). This model is the default.

Whole recordings. The privacy numbers cover the annotated human events. To check for speech the annotators did not mark, Whisper read all 8 s windows with no annotated human sound within 0.5 s (519 windows, 68 minutes). Ten windows produced two or more words in the unprocessed audio: eight were typical hallucinations over dog sounds and two were short phrases worth a listen. No substantial unannotated speech was found.

Out of domain. The detector was also run, unchanged and with the thresholds set on ELTE, on 40 synthetic 30 s scenes built from material it never saw: read English speech from 20 LibriSpeech speakers [12], 20 ESC-50 dog clips [13], rooms simulated with pyroomacoustics [14] (RT60 0.2–0.8 s) and pink noise, with three utterances and four dog clips per scene that partly overlap. Whisper small, scored against the reference transcripts, reads two or more correct words in 99% of the unprocessed utterances. After the AudioSet-only detector it still does in 11.7% (light) and 7.5% (standard); after the new detector in 1.7% and 0%, as after muting the exact utterances (0%). Silero VAD [15] leaves 11.7%. The price is dog sound: the new standard setting mutes 57% of the dog-only time in these scenes, against 47% for the AudioSet detector and 47% for the exact utterances, because foreign barks overlap the speech and are hard to tell from it. An earlier attempt to reach this privacy by training the AudioSet detector on synthetic speech-under-bark scenes needed 63–77% dog loss.

6 Keeping the dog under speech

227 s of dog sound lies under a human sound (223 calls, 178 of them pants); the standard setting mutes 184 s of it. With the detector's false alarms it mutes 12.1% of all dog sound, about 40% of it under a person.

Dog extractor. A network predicts a time–frequency mask $M \in [0, 1]$ on the 16 kHz spectrogram (512-point STFT, 16 ms hop) that keeps the dog: three convolution layers on the log power, concatenated per frame with the Whisper encoder frames of the same audio (projected to 128 values), a two-layer bidirectional GRU and a sigmoid output (1.2M parameters). It is trained to regress a privacy-weighted Wiener mask, $|D|^2/(|D|^2 + 10\,|S|^2)$, with D the dog and background and S the human sound, on mixtures of the lab's own audio: 4 s windows with dog calls and no person, plus real human-only events from the same participant group (30% LibriSpeech speech in simulated lab rooms), 3,000 per group. Two folds split the participants (odd/even numbers); each session is processed by the model of the other fold. On held-out mixtures it removes 6–12 dB of the human sound while turning the dog down by only 0.6–3.7 dB (dog SI-SDR 9–15 dB); the folds differ, with the remaining speech 20–22 dB below the dog in one and only 2–4 dB in the other. The same network without the Whisper frames removed only 4–6 dB: with $\sim$150 real human events per fold it learns the training voices, not speech.

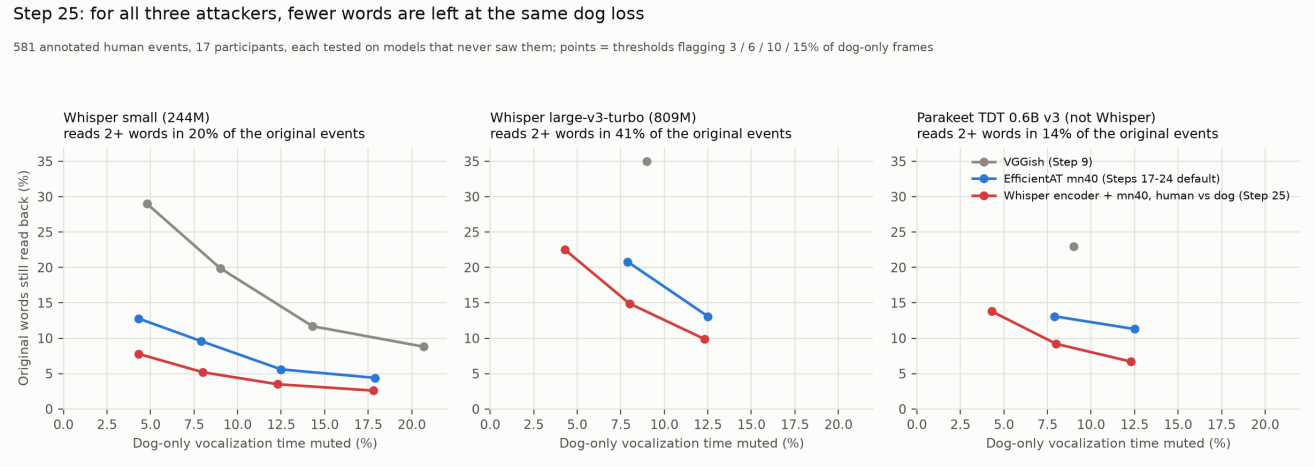


Figure 1: Original words read back after muting, for three attackers, against the dog-only vocalization time muted. Each line is one detector at 3/6/10/15% of the dog-only frames flagged (turbo and Parakeet were run at fewer points). On background audio, the share of an event's words that turn up is 0.0% (Whisper small, Parakeet) and 1.6% (turbo).

Table 4: Keeping the dog inside the muted time (standard setting, mask M^3). Words read back by Whisper small / Parakeet / turbo (%), and the share of muted dog time that comes back: dog calls under a person, and dog-only calls the detector flagged. Last column: all dog time muted. Tone: room tone 10 dB above the room's own; *all*: extractor everywhere, tone where the 5% router hears no dog. *Nothing is muted, but quiet calls lie near the level of the added tone.

inside muted time	S / P / T	under	flagged	lost
silence	4.4 / 7.9 / 17.0	0	0	12.1
extractor everywhere	5.8 / 16.9 / –	100	100	–
router 2% (safe)	4.4 / 7.9 / 17.9	18	19	**9.8**
router 5% (more)	4.4 / 10.4 / 18.0	32	31	**8.2**
+ mics, 10%	4.4 / 11.7 / –	52	49	6.0
everywhere + tone	4.4 / 11.3 / 17.6	100	100	0*
all	4.4 / 13.8 / 18.4	100	100	**0***

Where to use it. Inside the muted time of the real sessions, replacing silence by the extractor's output everywhere brings speech back (Whisper small: 2.8% of the events readable instead of 0.9%; Parakeet reads back 16.9% of its words instead of 7.9%). Using it only inside the annotated dog calls (an oracle) keeps privacy close to muting (1.2% and 12.5%): where the dog vocalizes, it covers what is left of the speech. A *router* therefore decides where a dog is calling: the human detector's features, trained on frames with vs without a dog call (frames with a person weighted 5×), leave-one-dog-out. Inside human frames it separates dog+person from person-only with AUC 0.78 (0.81 with the clip-on microphone cues). Its threshold is the share of person-only frames it lets through.

Result. Table 4: with 2% of person-only frames let through, Whisper small reads exactly the same events and words as after muting, Parakeet reads back the same share of words (4 events become readable, 6 stop being readable), and turbo reads back 17.9% instead of 17.0% of its words (the gained words are fillers such as "you", "so"); a fifth of the muted dog sound comes back and the share of all dog sound lost falls from 12.1% to 9.8% (16.4% for the AudioSet-only detector). At 5%, a third comes back (8.2% lost); Whisper small reads the same words, Parakeet 10.4% instead of 7.9% (mostly laughter), and turbo reads one short phrase in full that muting removes. With 5–54 readable events per attacker, "no measurable change" is not a proof of equal privacy. The table's "silence" row comes from its own transcription run and differs slightly from Table 2. With the clip-on microphone cues the router lets more dog through at the same rate (38% of the dog under a person at 5%). The kept dog is the extractor's output: 16 kHz, a few dB quieter, with the speech removed (Figure 2). The mask exponent 3 was chosen after exponent 1 had been scored on the same events (exponent 1, router 2%: Parakeet reads back 10.6% of its words), so this setting is slightly optimistic; the router's threshold is an average, and per participant the safe setting lets through 0–15% of the person-only frames.

Room tone. Why does the extractor's output bring speech back when used everywhere? On clean unseen voices it removes 14 dB of the speech (20 dB with M^3), but the attackers normalize the level before reading, and in the quiet lab the leftover voice is still the loudest thing in the output: person-only speech is about 15 dB above the room tone, the extractor's output during it 4–6 dB. We therefore add, inside the processed time, stationary noise with the spectrum of the recording's own room tone (the quietest 20% of the frames outside the processed time), 10 dB above it. With the extractor everywhere, Whisper small reads back the same words as after muting and Parakeet 11.3% (7.9%); an adaptive attacker that first enhances the audio with DeepFilterNet3 [16] before Parakeet reads 15.6% (10.0% after muting). The extra events contain laughter, "mm-hmm" and "thank you"—no new words; turbo reads 17.6% (17.0%). Adding the noise only where the router hears no dog ("all") keeps the routed dog clean: Parakeet reads back 13.8% (+5.8 points, 95% CI 1.9–11.8 over participants, all laughter, backchannels and "thank you"), DeepFilterNet+Parakeet 16.1%, turbo 18.4% (+1.4, CI −0.4–3.3; six events gain words, against four with the router alone). No dog time is muted; of the 432 annotated calls in the processed time, 34% are 6 dB or more above the added noise and 11% get no noise, but pants, the quietest calls (13 dB above the room tone), come back at about the noise level (median within 1 dB). A noise level that follows the power the extractor removed kept the dog

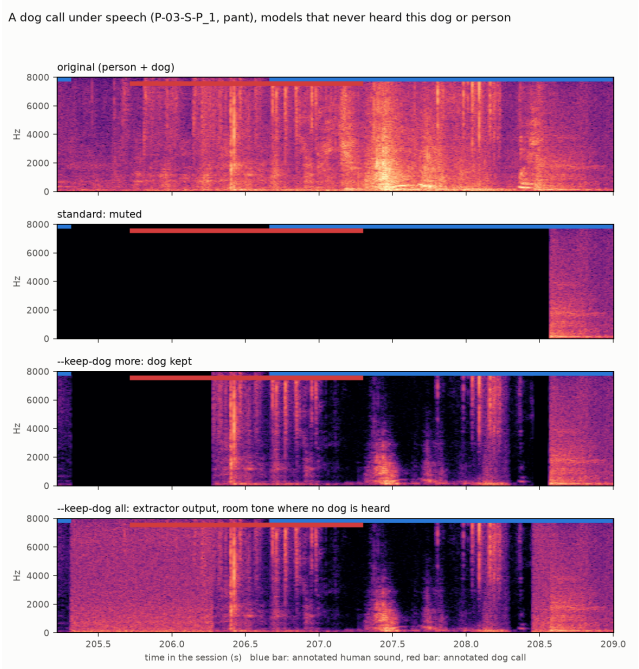


Figure 2: A pant under speech from a dog and person the models never heard: original, muted, dog kept where the router (5%) hears a dog, and the extractor's output everywhere with room tone where the router hears no dog. Blue: annotated human sound; red: annotated dog call.

more audible but let turbo read 5.5 points more, and was dropped.

7 What did not help

- *Subtracting instead of muting.* Where a dog overlaps the speech, subtracting a speech estimate (DTLN [17], or GTCRN [18] fine-tuned on this data) instead of muting keeps some dog sound, but at equal dog loss it never left fewer readable words than muting.
- *Three microphones.* Pooling the training data of all microphones and taking the maximum probability over them did not beat a single microphone with the same features.
- *20 ms decisions.* About 40% of the dog sound lost lies within 1 s of a human sound, which suggested finer boundaries. A classifier on every 20 ms Whisper frame (with $\pm$0.06/0.5/1.5 s context, trained against all other frames) muted 68% of the human time at 3.6% dog loss, against 73% at 4.5% for the 0.24 s detector with the same input and objective (77% at 7.1% vs 81% at 7.9%); interpolating the 0.24 s probabilities to 20 ms with tighter padding changed nothing.
- *Dog extractors from the spectrogram alone.* A privacy-weighted signal loss removed 19 dB of speech but also turned the dog down by 9 dB; mask regression on the spectrogram (and with the clip-on microphone's level difference) removed only 4–6 dB of held-out speech.
- *Better extractors and routers.* 780 synthetic voices (Piper and VITS text-to-speech, Hungarian and English, split by speaker between the folds), a loss on the leaked speech in dB, and a U-Net-style frequency decoder each moved the extractor along the same speech-vs-dog trade-off (the privacy-weighted target mask itself would remove 31 dB). Describing the extractor's output with the detector's networks did not make the router better (AUC 0.74–0.80).
- *Other inputs and labels.* Training on speech-only labels (events where Whisper reads words) instead of all human sounds, normalizing the scores per recording, adding neighbouring frames or the standard deviation of the Whisper frames were all equal or worse. Subtracting a separate dog detector's score helped only with a tuned weight, and less than changing the training objective.

8 Discussion and limitations

The largest gains came from the input (a speech encoder) and from the training objective (human vs dog), not from the classifier or the time resolution. The detector is a linear model on frozen features, trained on 16 dogs in a few minutes on a laptop, and the full anonymizer, including both frozen networks, runs at about 8$\times$ real time on two CPU cores.

The evaluation has limits. All real data come from one lab, one room and one kind of interaction; the synthetic out-of-domain test covers other speakers, dogs and rooms only in simulation. Privacy is measured by automatic recognizers on the annotated events; a human listening test, of the remaining readable events and of the kept dog sound under speech, is still to be done (a pack of 12 clips is prepared). The threshold fixes the dog cost on average: at the 6% setting the share of dog-only frames flagged ranges from 0% to 19% between dogs. Handler identity is not in the metadata, so if one person played with several dogs, the split is disjoint in dogs but not in speakers. The inputs, objective and settings were chosen over several rounds on the same 17 folds, which makes the best configuration somewhat optimistic; the out-of-domain test is the guard against this. The kept dog sound under speech is a processed signal at 16 kHz, not the original recording, and the router finds only a third to a half of it at an acceptable privacy cost; with room tone all of it comes back, but quiet calls under a noise floor. The extractor passes laughter.

9 Conclusion

For anonymizing dog–human play recordings, a frozen speech encoder combined with AudioSet class scores, trained on human versus dog sounds with every call type weighted equally, reduces the readable speech to the level a recognizer reaches on background noise while losing a third less dog vocalization than an AudioSet-only detector, and a Whisper-conditioned dog extractor, gated by a dog router, brings back part of the dog sound under speech at no measurable privacy cost; with the recording's own room tone masking what is left of the voice, no dog sound needs to be silenced at all. The code, the evaluation and a one-command anonymizer that keeps the original file format are available at [repository URL once public].

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